
POURIA EBRAHIMNEZHAD

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PROFESSIONAL SUMMARY

Lead Data Scientist with 10+ years across Australia's telecommunications and healthcare sectors. Currently leading GenAI initiatives at **I-MED Radiology**, Australia's largest medical imaging network — building agentic AI assistants for customer-facing operations on AWS Bedrock with Claude. Previously led the ML team behind **Telstra's customer-experience program**, delivering **\$4.2M in annual savings** through churn modelling and applying LLM-based topic modelling to customer segmentation. I work across the technical and product sides of AI — comfortable in the architecture and in the room with executive stakeholders

KEY ACHIEVEMENTS

- Lead production deployment of an agentic AI assistant (chat and voice) at I-MED Radiology, handling 17K+ monthly customer enquiries with 6× cost efficiency per interaction and on track to deflect 40% of voice traffic to digital self-service.
- Led development of ML propensity models driving Telstra's CIQ initiative, resulting in enterprise-wide adoption through integrated dashboards, enabling data-driven store operations across 15 locations, and informing strategic decisions for network investment and CX improvement programs
- Pioneered innovative application of LLM-based topic modeling to analyze emerging customer experience drivers and trends.
- Delivered \$4.2M annual savings through development and deployment of multiple ML churn prediction models, enabling targeted retention campaigns and uncovering key churn drivers
- Uncovered novel insights into customer churn behavior and key drivers in assurance, leveraging data from diverse sources including digital interactions and assurance interactions.
- Developed strategic data visualization suite integrating external property data (CoreLogic) to identify and retain moving customers, improving move retention rates through enhanced targeting and messaging
- Received Director's Accolade for optimizing customer support processes through successful integration of multiple products across assurance and support teams, improving fault resolution times.

DEMONSTRATED RELEVANT SKILLS

Machine Learning / Data Modelling

- Advanced Supervised Learning: classification and regression using GBM, XGBoost, LightGBM, Random Forest
- Natural Language Processing: LLM integration, topic modeling, sentiment analysis
- Feature Engineering & Selection: Complex feature development, dimensionality reduction
- Time Series Analysis: Forecasting, trend analysis, seasonal decomposition, ARIMA, Prophet
- Deep learning: LSTM, RNN, Transformers, Keras, TensorFlow
- Specialized Techniques: Imbalanced dataset handling, propensity modeling

- Model Development: End-to-end ML pipeline development, advanced model evaluation, hyperparameter tuning
- Exploratory Data Analysis: Statistical analysis, data visualization, pattern recognition

GenAI / LLM Engineering

- LLM Integration: OpenAI API implementation, streaming architectures, Ollama deployment
- HuggingFace pipelines, tokenizers and models, combine frontier and open-source models
- RAG: Use Vector embeddings, VectorDB, LangChain, LangGraph for context enrichment and retrieval
- Foundation Model Platforms: AWS Bedrock with Anthropic Claude, OpenAI API, Ollama for local deployment
- Agentic Systems: tool use, multi-step reasoning, agent orchestration with Strands and LangGraph, Model Context Protocol (MCP)
- Conversational & Voice AI: production chat and voice agents, telephony integration evaluated on Amazon Lex and Amazon Nova
- LLM Evaluation: benchmarking foundation models for domain-specific tasks (e.g. clinical report generation), accuracy and operational fit assessment

Data Engineering & Analytics

- Programming Languages & Tools:
 - Python: Advanced ML libraries, data manipulation, custom model development
 - SQL: Complex queries and feature engineering across enterprise databases (Teradata, AWS Redshift, Azure)
 - R: Statistical analysis, R-Shiny dashboards, data modeling
- MLOps & CI/CD:
 - Version Control: Git, GitHub for collaborative development
 - Model Monitoring: Performance tracking, automated testing
 - Production Deployment: Model versioning, automated validation
 - ETL processes across multiple data sources
- Data Visualization & BI
 - Enterprise Tools: PowerBI, Tableau, Splunk for reporting and dashboards
 - Programming-based: Advanced visualizations using Python (Matplotlib, Seaborn) and R
 - Interactive Development: R-Shiny applications, dynamic reporting
 - Advanced Excel for business analysis and reporting

Cloud Technologies

- Advanced in AWS Sagemaker, Athena, S3, Cloudwatch
- Experienced in Microsoft Azure ML
- Deployment of Splunk on AWS cloud
- Intermediate in Linux, navigating file system, managing files and directories

EMPLOYMENT HISTORY

Lead Data Scientist | I-MED Radiology - Melbourne (August 2025 – Current)

Workstream - GenAI for operational efficiency – Project Imogen

- Lead the design, architecture, and production deployment of an **agentic AI assistant** (chat and voice) handling **17K+ monthly customer enquiries** across billing, clinical information, procedure details, pricing, and booking workflows — delivering **6× cost efficiency per interaction** and on track to **deflect 40% of voice traffic to digital self-service**.

- Architected a **multi-framework agent stack on AWS Bedrock with Anthropic Claude: Strands** for orchestration, **RAG** over internal knowledge bases via vector search, and a voice layer evaluating **Amazon Lex and Amazon Nova** for telephony integration.
- Drive the GenAI workstream **end-to-end** — from use-case prioritisation and stakeholder workshops through technical design, evaluation, and production rollout — partnering with **clinical operations, IT, and customer service leadership**.
- Lead **LLM benchmarking and evaluation** for clinical report generation across **ultrasound and DEXA worksheets** — assessing accuracy and operational fit for production use.
- Scope and shape the **annual Data Science roadmap and team strategy** at I-MED, defining GenAI use-case prioritisation across the operational efficiency portfolio.

Senior Data Scientist | Telstra - Melbourne (July 2023 – May 2025)

Project - Customer Experience Management – USE NPS Customer-IQ

- Lead the **development of ML models** for predicting **customer experience**.
- Lead the data transformation including **feature engineering**, required for model training and optimization
- Lead the development of diagnostics for assessing **model performance** and analysis
- Lead the implementation of segmentation models Using **LLM** capabilities for **topic/sentiment analysis**
- **Lead** the team in **ideation** and problem solving using **data insights** to inform **commercial and marketing strategies**
- liaison with project leads and **project management** for ideation and scoping
- Lead and mentor junior data scientist with technical project delivery and lifecycle of data science projects

Data Scientist | Telstra – Melbourne (Sep 2021 - July 2023)

Project - Customer Base Management Personalisation & AI

- Building various **Machine learning** and **predictive models** in **production**
- **Running Workshops** with **business** stakeholders and marketing teams
- Performing **exploratory data analysis** on multiple business questions using various data sources, providing **insights** to highlight **opportunities** to improve **customer experience** and uplift retention.
- **ML Ops** monitoring the performance of **in production models** and managing the **data pipeline**
- Designing experiments on marketing campaigns and Performing PIR

Data Scientist | Telstra - Melbourne (May 2019 – Sep 2021)

TEAM – Sports Media Analytics & Advertising COE

- Design and Build **reporting** and **visualisation dashboards** using Tableau, Splunk and PowerBI using aggregated data sets for media product teams
- Perform **analysis** and quantification of customer and **user experience** for live **sports streaming** apps including AFL, NRL, Netball and AFLW
- develop proof-of-concept for **churn model** for media products life cycle management
- Support various business units with **data engineering** and data onboarding solutions using **Splunk**
- Develop **Splunk dashboards** and **alarming** for media operation teams

Technical Expert | Telstra - Melbourne (Oct 2013 to May 2019)

Network and Services Operations - Customer Assurance

- SME for customer incident management and problem management across various services and products

- Conduct root cause analysis and technical deep dive investigation and analysis of network faults
- Liaise with network operations and engineering teams to investigate and identify complex network problems
- Lead the team in operational enablement including tools and process development and improvement

EDUCATION & CERTIFICATION

Graduate Diploma: Data Science, 2021
Monash University - Melbourne, VIC

Master of Science: Telecommunications and System Engineering, 2010
Shiraz University – Shiraz, Iran

Bachelor of Science: Electrical Engineering - Telecommunications, 2005
Isfahan University of Technology – Isfahan, Iran

Certification and Courses:

- Microsoft AI-900 AI fundamentals Certified
- LLM Engineering - Udemy
- AWS Certified Machine Learning Speciality - Udemy
- Transfer Learning for NLP with TensorFlow Hub - Coursera